NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Empowering AI in Sciences Through Common Data Preparation Principles

Melanie HERSCHEL
Associate Professor
College of Computing and Data Science

AI for Sciences Winter School 2025

Agenda

- Motivation and Introduction
- Common data preparation tasks
- Discussion phase: Example applications
- Wrap-up + Q&A

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Data Preparation ?!?
What is it, and why bother ?

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AI starts long before the model

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NEWS | 09 October 2024

Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures

This year's prize celebrates computational tools that have transformed biology and have the potential to revolutionize drug discovery.

By Ewen Callaway

For the first time – and probably not the last – a scientific breakthrough enabled by artificial intelligence (AI) has been recognized with a Nobel prize. The 2024 chemistry Nobel was awarded to John Jumper and Demis Hassabis at Google DeepMind in London, for developing a game-changing AI tool for predicting protein structures called AlphaFold, and David Baker, at the University of Washington in Seattle, for his work on computational protein design, which has been bolstered by AI in the past few years.

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Article | Open access | Published: 16 January 2025

A generative model for inorganic materials design

Claudio Zeni, Robert Pinsler, Daniel Zügner, Andrew Fowler, Matthew Horton, Xiang Fu, Zilong Wang, Aliaksandra Shysheya, Jonathan Crabbé, Shoko Ueda, Roberto Sordillo, Lixin Sun, Jake Smith, Bichlien Nguyen, Hannes Schulz, Sarah Lewis, Chin-Wei Huang, Ziheng Lu, Yichi Zhou, Han Yang, Hongxia Hao, Jielan Li, Chunlei Yang, Wenjie Li, ... Tian Xie
+ Show authors

Nature **639**, 624–632 (2025) | Cite this article

178k Accesses | **303** Citations | **258** Altmetric | Metrics

Abstract

The design of functional materials with desired properties is essential in driving technological advances in areas such as energy storage, catalysis and carbon capture[1,2,3]. Generative models accelerate materials design by directly generating new materials given desired property constraints, but current methods have a low success rate in proposing stable crystals or can satisfy only a limited set of property constraints[4,5,6,7,8,9,10,11]. Here we present MatterGen, a model that generates stable, diverse inorganic materials across the periodic table and can further be fine-tuned to steer the generation towards a broad range of property constraints. Compared with previous generative models[4,12],

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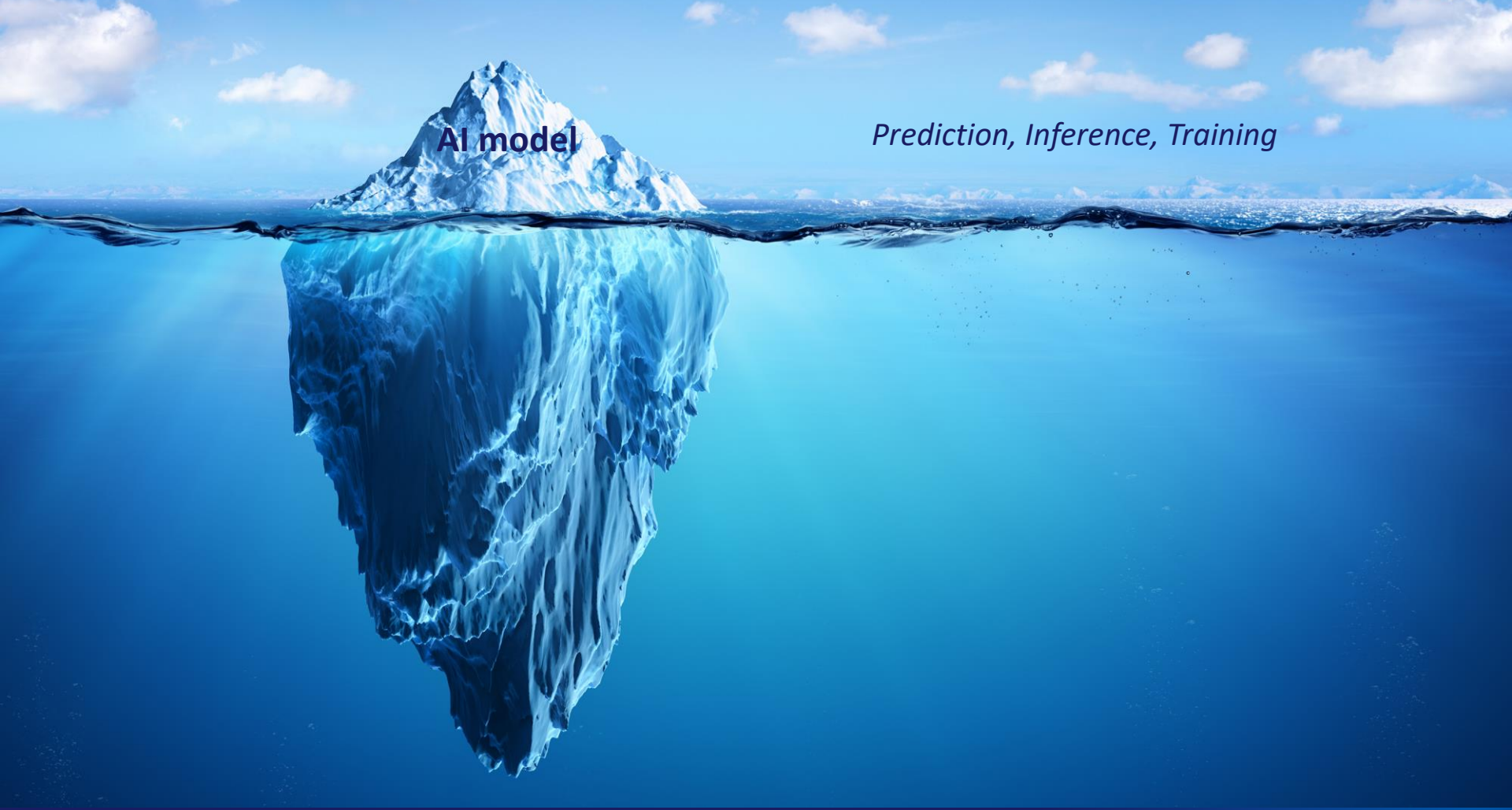

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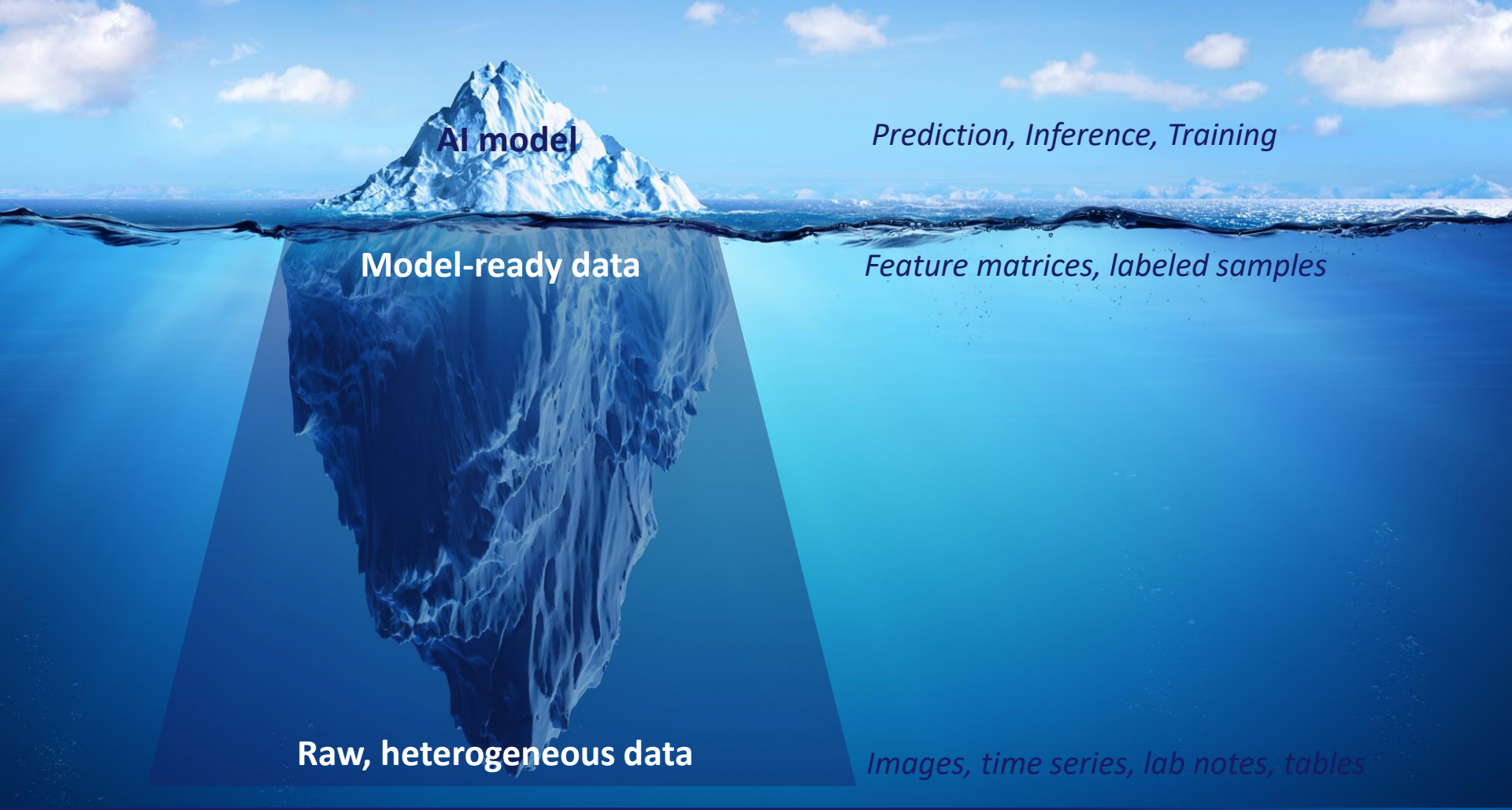

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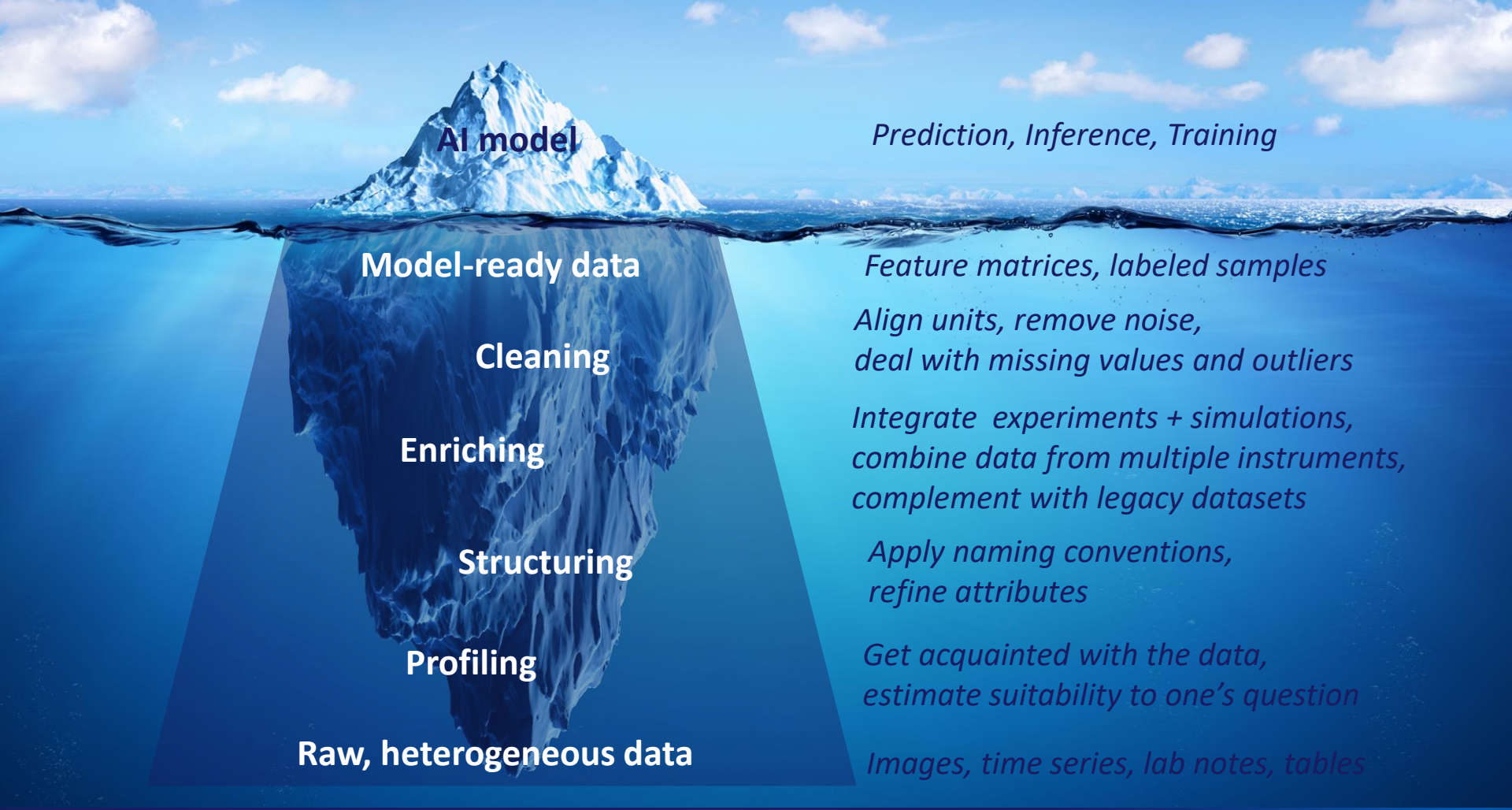

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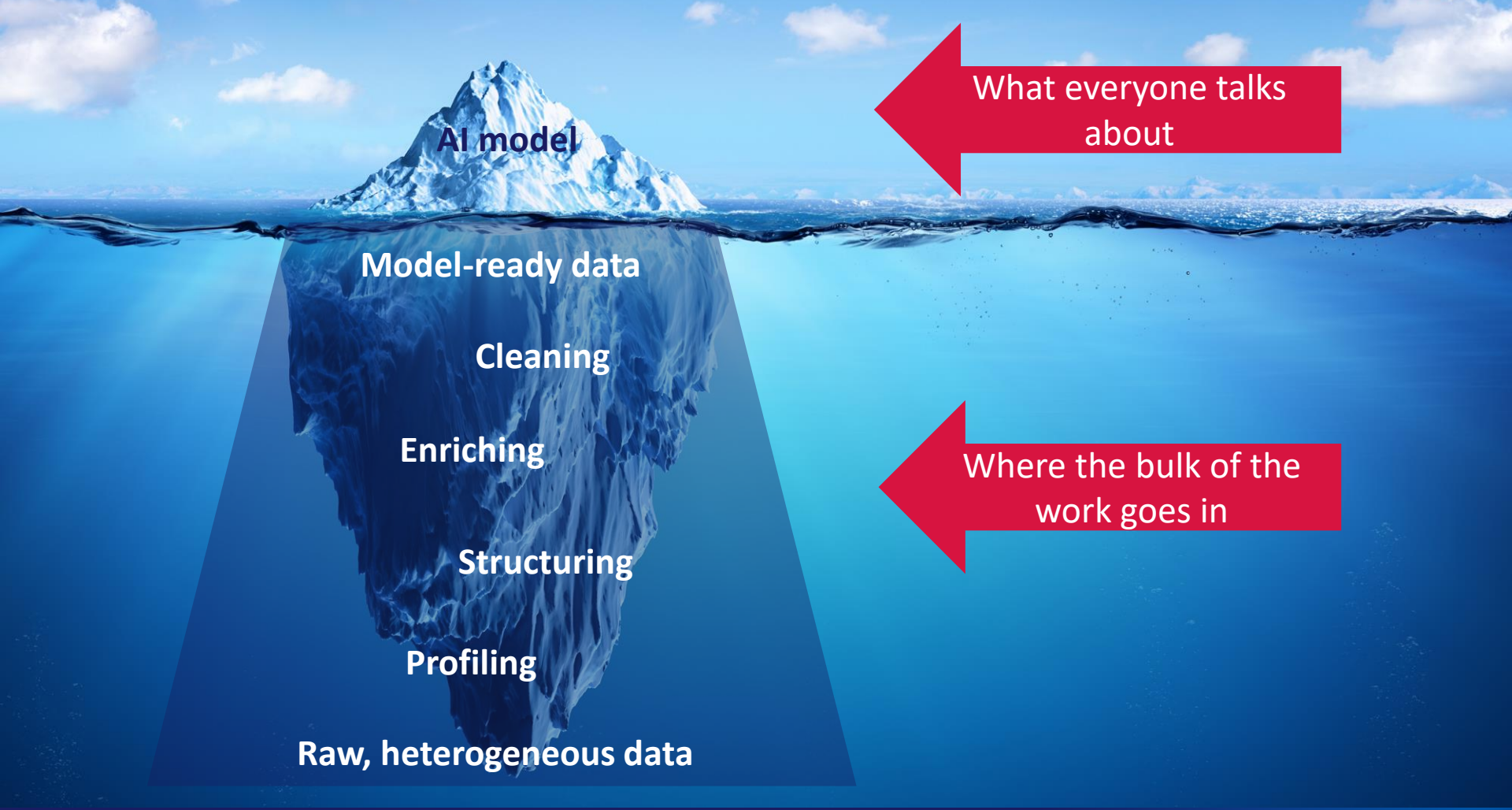

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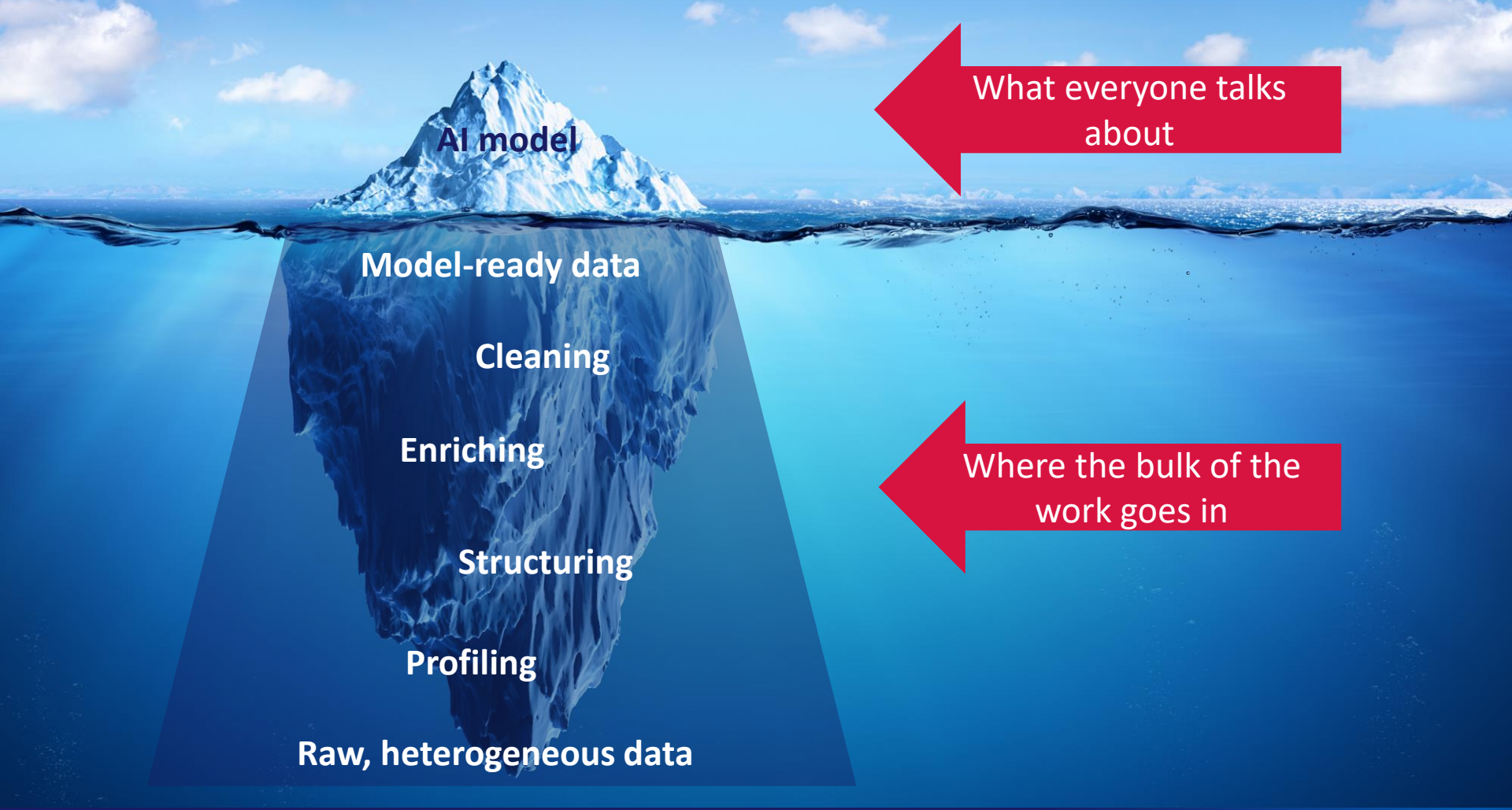

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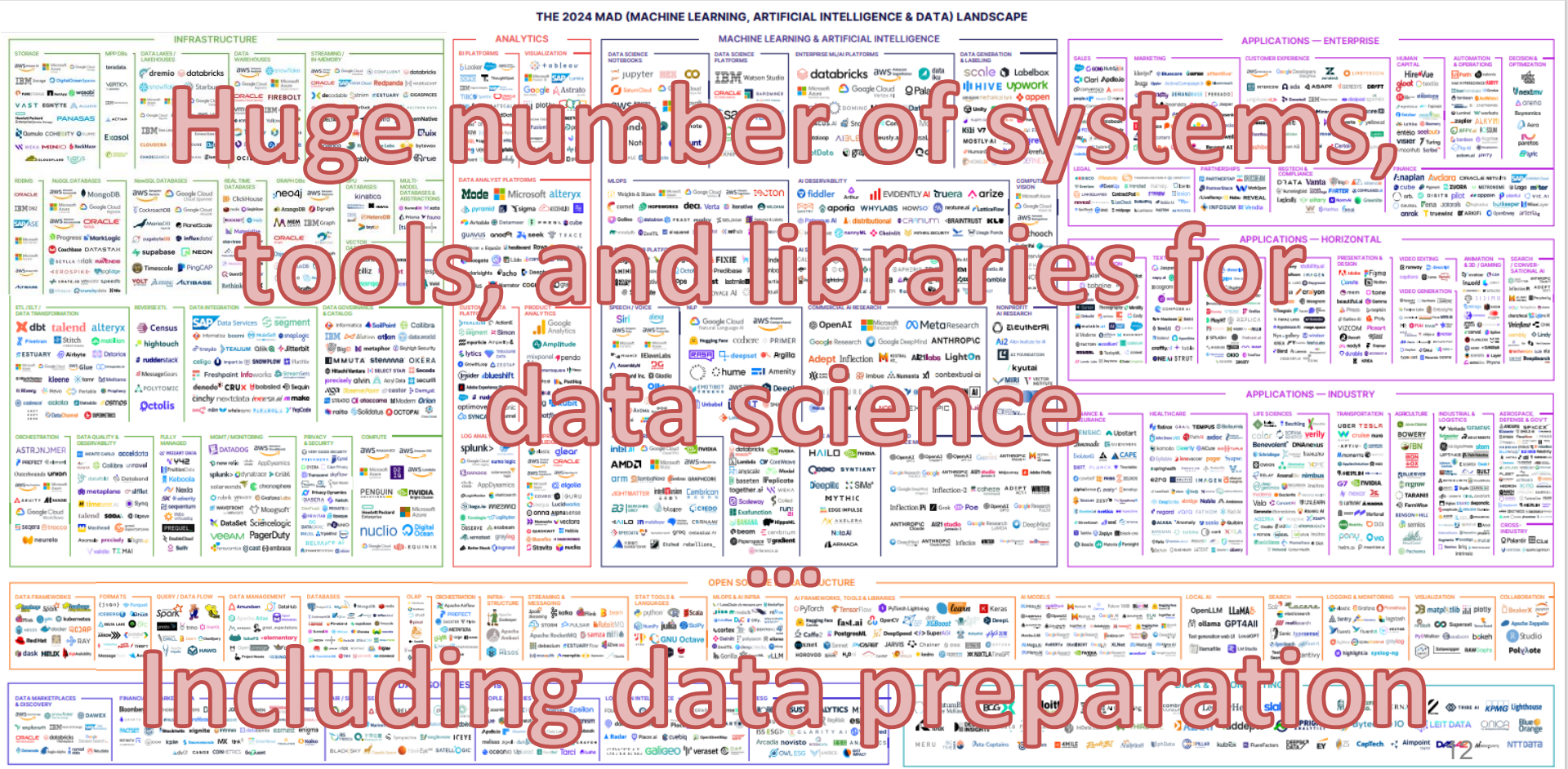

Huge number of systems, tools, and libraries for data science …

Including data preparation

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Objectives of today's talk

- Introduce you to the often overlooked available support for data preparation.
  - In terms of existing software
  - In terms of scientific collaboration
- Highlight common steps in data preparation
  - Available in systems/tools/libraries
  - Subject to further research

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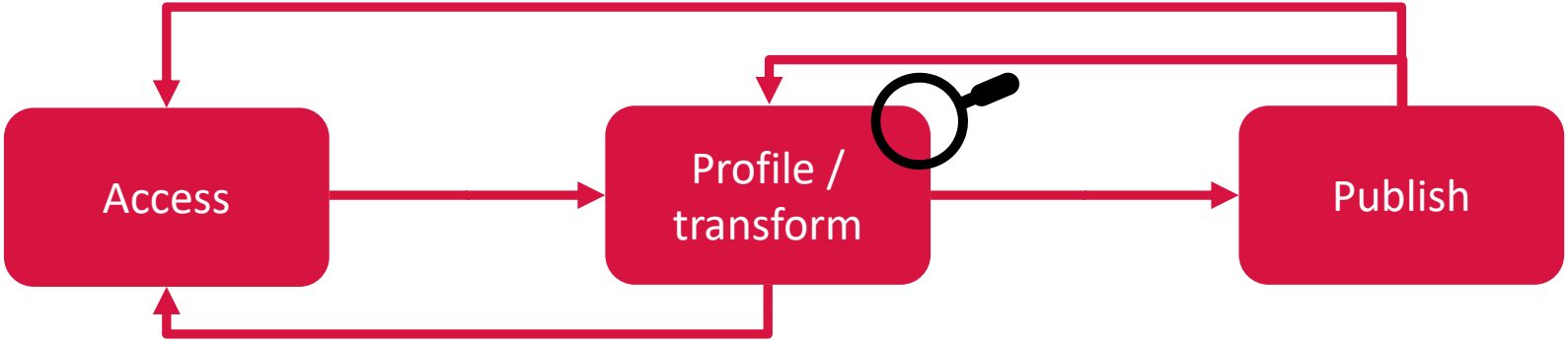
Data preparation process

Data preparation is the process of **converting raw** data into a **form** that allows to extract **value** from data.



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Subtasks of central component

Data Profiling	Data Structuring
Data Enriching	Data Cleaning

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Data Profiling

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Intuitive data profiling introduction

Data profiling is the set of activities and processes to determine the metadata about a given dataset.

- Examine and analyze data to create metadata (data about data).
- Metadata represent a summary of various characteristics of the data.
- **These summaries help to assess usefulness and quality of data.**

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Simple examples:

- Number of rows
- Min and max values
- Number of missing values
- Outliers
- Data type
- Data format violations
- Histograms
- …

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Running example for single column profiling

- Relies on ydata Profiling
- Integrates into Python Notebooks
- Example profiling of the Titanic Dataset
- Full ydata Profiling report for this dataset available on the ydata examples page.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103
13	0	3	Saundercock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05	

Excerpt of Titanic Dataset

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Cardinalities

Cardinalities for the Age Attribute

Number of distinct values
Number of missing values
MIN, MAX, MEAN Age
…

Age

Real number (ℝ)

Missing

Age

Distinct	88	**Minimum**	0.42
Distinct (%)	12.3%	**Maximum**	80
Missing	177	**Zeros**	0
Missing (%)	19.9%	**Zeros (%)**	0.0%
Infinite	0	**Negative**	0
Infinite (%)	0.0%	**Negative (%)**	0.0%
Mean	29.699118	**Memory size**	7.1 KiB

High number of missing values may be a problem for age-related analysis.

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Data types and value patterns

Age

Real number (ℝ)

Missing

Inferred from CSV

Distinct	
Distinct (%)	
Missing	
Missing (%)	
Infinite	
Infinite (%)	
Mean	

Embarked

Categorical

Inferred from CSV

Distinct	3
	0.42
	80
	0

Variable types impact how the attributes are to be handled prior / during analysis.

Overview | Alerts 10 | Reproduction

Summary of detected types

Dataset statistics

Number of variables	12
Number of observations	891
Missing cells	866
Missing cells (%)	8.1%

Variable types

Numeric	5
Categorical	4
Text	3

Limitations of standard tools

- Focus on tabular data (but Data Structuring can be used to map data into structured data)
- Support single-column profiling (computationally easy)
- Multi-column profiling requires advanced algorithms
  - Active area of research
  - Discovery of different types of data dependencies

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Data Structuring

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Overview of structuring transformations

- Structuring involves **changing the structure and granularity** of a dataset (i.e., changes form and schema of the data).
- Structuring transformations create **new data based on data already present** in the dataset.
- Knowing the common classes of operations helps to make **effective use of tools** designed to alleviate the structuring effort and that come with **built-in documentation**.

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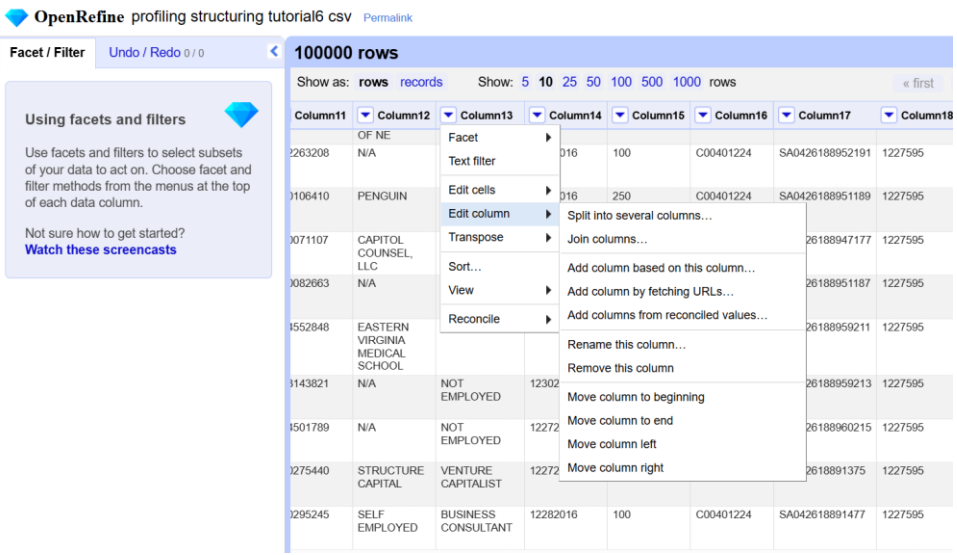

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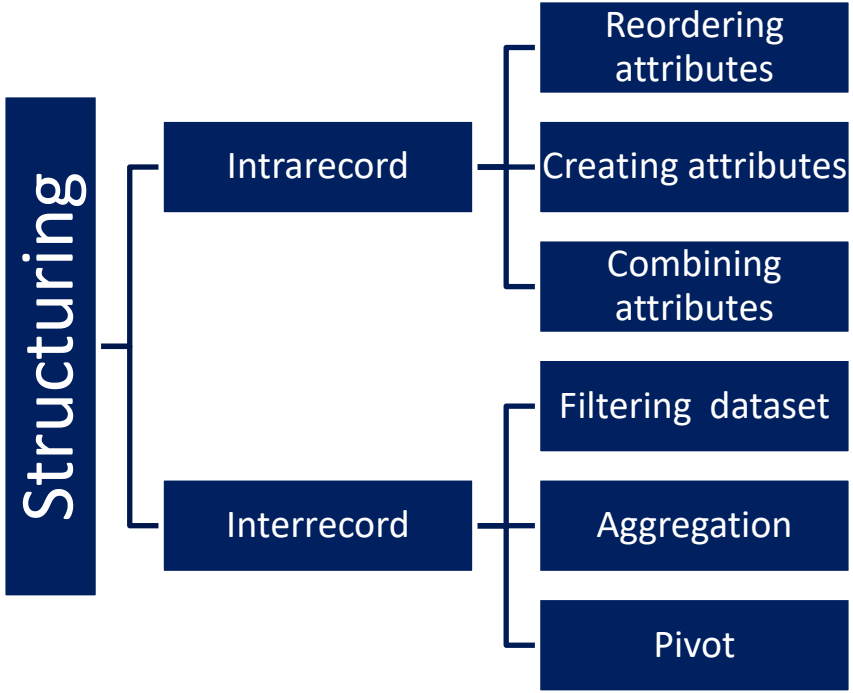
Overview of structuring transformations



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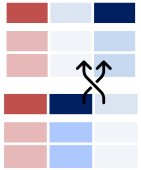
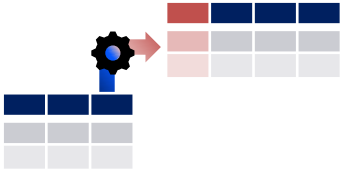
Intrarecord structuring



Reordering attributes

Attributes and values remain unchanged



Creating attributes

Attributes are added with values **extracted** from existing attributes

Combining attributes

Attributes are **merged** into a single attribute

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Extracting Values

- Extraction involves **creating a new column** (i.e., attribute and values for this new attribute) **from an existing column**.
- Three common extraction approaches:
  - Positional extraction
  - Pattern extraction
  - Complex structure extraction

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Extracting values – Pattern extraction

- Uses rules to describe the sequence of characters or patterns to be extracted.
- Such patterns can frequently be specified by regular expressions (regex).

- Some regex notation
  - \d matches a digit
  - \s matches a whitespace
  - * for 0 or more occurrences
  - ? For 0 or one occurrence
  - | either or
  - [0-9] accepted range of values
- Full reference: Python RegEx

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Extracting values – Pattern extraction

Column20
P/R DEDUCTION ($97.85 MONTHLY)
P/R DEDUCTION ($25.00 MONTHLY)
P/R DEDUCTION ($75.00 MONTHLY)
P/R DEDUCTION ($36.55 MONTHLY)
P/R DEDUCTION ($20.00 MONTHLY)
P/R DEDUCTION ($45.00 MONTHLY)
P/R DEDUCTION ($20.00 MONTHLY)
P/R DEDUCTION ($60.00 MONTHLY)
P/R DEDUCTION ($20.00 MONTHLY)
P/R DEDUCTION ($25.00 MONTHLY)
P/R DEDUCTION ($25.00 MONTHLY)
P/R DEDUCTION ($25.00 MONTHLY)
P/R DEDUCTION ($20.00 MONTHLY)
P/R DEDUCTION ($25.00 MONTHLY)
P/R DEDUCTION ($15.00 SEMI-MONTHLY)
P/R DEDUCTION ($30.00 MONTHLY)
P/R DEDUCTION ($42.11 SEMI-MONTHLY)

- We want to create a new column MONTHLY_CONTRIBUTION
- Needs extraction of the digits + dot + digits between the $-sign and the keyword MONTHLY.

Regular expression:
\$\d+.?\d*\sMONTHLY

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Extracting values – Pattern extraction

Regular expression:
\$\d+.?\d*\sMONTHLY

- Regular expressions integrated in `re` Python library
- search() returns all substrings of text that match the expression (can be more than one).

```python
import re

def extract_contribution(txt):
  # Search for the pattern
  # $ + digits + . (optional) + digits (optional) + space + MONTHLY
  match = re.search(r"\$\d+.?\d*\sMONTHLY", str(txt))
  if match:
    # Remove MONTHLY suffix and $ prefix
    return float(match.group(0).split(' ')[0][1:])
  else:
    return np.NaN

df['MONTHLY_CONTRIBUTION'] = df['Column20'].apply(extract_contribution)
print(
      df[['Column20', 'MONTHLY_CONTRIBUTION']].
        sort_values(by='MONTHLY_CONTRIBUTION', ascending=False).head(100
      )
```

```
                              Column20  MONTHLY_CONTRIBUTION
3454   P/R DEDUCTION ($437.50 MONTHLY)                437.50
16316  P/R DEDUCTION ($416.66 MONTHLY)                416.66
26381  P/R DEDUCTION ($416.66 MONTHLY)                416.66
14576                ($416.66 MONTHLY)                416.66
```


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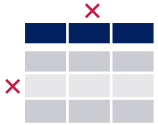
Interrecord structuring

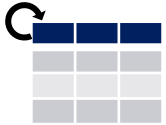


Filtering

Removal of columns or rows.

Aggregation

Attributes are added with values **extracted** from existing attributes



Pivot

Attributes are **merged** into a single attribute

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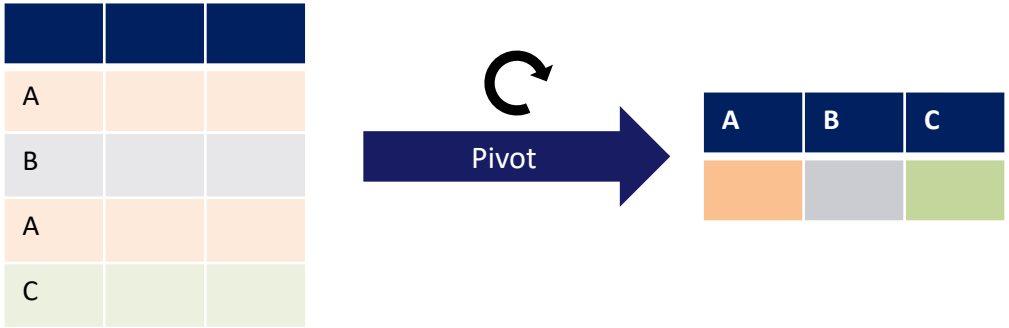
Row-to-column pivot

- Output records are based on multiple input records
- Values of output record attributes are based on some aggregation of input record attribute values.
- Frequently combined with aggregation, such that input records can contribute to multiple output records.



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Pivot – Example

- Six distinct values of Column7 identify different source types of contributions.
- Contributions made from various US states.
- Aggregation + pivot to count the number of contributions per state (i.e., Column10 remains) and source type (pivots from row values to columns)

```
#Create data grouped by source type and state
grouped = df.groupby(
    ['Column7','Column10'])['Column1'].agg(['count'])

#Pivot grouped to get Column7 as columns
pivot_table = grouped.pivot_table(
      index='Column10',
      columns='Column7',
      values='count',
      aggfunc='sum',
      fill_value=0
    )
print(pivot_table)
```

Column7	CAN	CCM	COM	IND	ORG	PAC
Column10						
AA	0	0	0	1	0	0
AE	0	0	0	2	0	0
AK	0	0	0	402	0	0
AL	0	0	0	2242	1	0
AR	0	0	0	1683	0	0
AZ	0	0	0	2278	0	0
CA	2	0	0	10301	12	0

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OpenRefine profiling structuring tutorial6 csv Permalink

Facet / Filter | Undo / Redo 0 / 0

Using facets and filters

Use facets and filters to select subsets of your data to act on. Choose facet and filter methods from the menus at the top of each data column.

Not sure how to get started?
Watch these screencasts

100000 rows

Show as: **rows** records Show: 5 **10** 25 50 100 500 1000 rows « first

Column11	Column12	Column13	Column14	Column15	Column16	Column17	Column18
	OF NE						
2263208	N/A		016	100	C00401224	SA0426188952191	1227595
0106410	PENGUIN		016	250	C00401224	SA0426188951189	1227595
0071107	CAPITOL COUNSEL, LLC					26188947177	1227595
0082663	N/A					26188951187	1227595
4552848	EASTERN VIRGINIA MEDICAL SCHOOL					26188959211	1227595
3143821	N/A	NOT EMPLOYED	12302			26188959213	1227595
4501789	N/A	NOT EMPLOYED	12272			26188960215	1227595
0275440	STRUCTURE CAPITAL	VENTURE CAPITALIST	12272			2618891375	1227595
0295245	SELF EMPLOYED	BUSINESS CONSULTANT	12282016	100	C00401224	SA042618891477	1227595

- Facet
- Text filter
- Edit cells
- Edit column
  - Split into several columns…
  - Join columns…
  - Add column based on this column…
  - Add column by fetching URLs…
  - Add columns from reconciled values…
  - Rename this column…
  - Remove this column
  - Move column to beginning
  - Move column to end
  - Move column left
  - Move column right
- Transpose
- Sort…
- View
- Reconcile

Data Enriching

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Overview of enriching

- Results in the net **addition of information** to a dataset.
  - Inserts additional records or attributes from other related datasets.
  - Use formulas to calculate new fields.
- Three **primary types** of enriching transformations:
  - Unions
  - Joins
  - Deriving new fields

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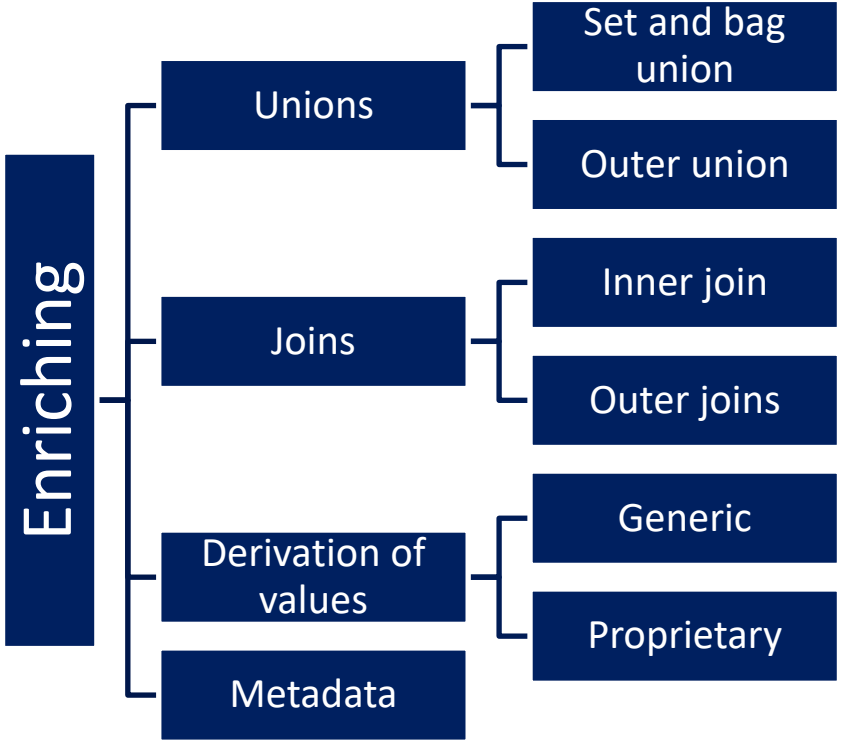
Overview of enriching



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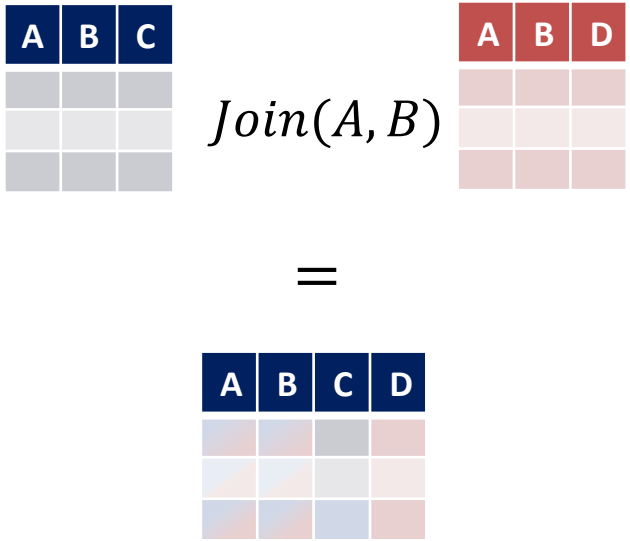
Joins

- Linking records from one dataset R to records from another dataset S.
- Linking based on a condition on (join) key attributes, e.g.,
  - Exact match of key values (most common)
  - Approximate (= fuzzy) match of key values (e.g., to account for data quality issues)



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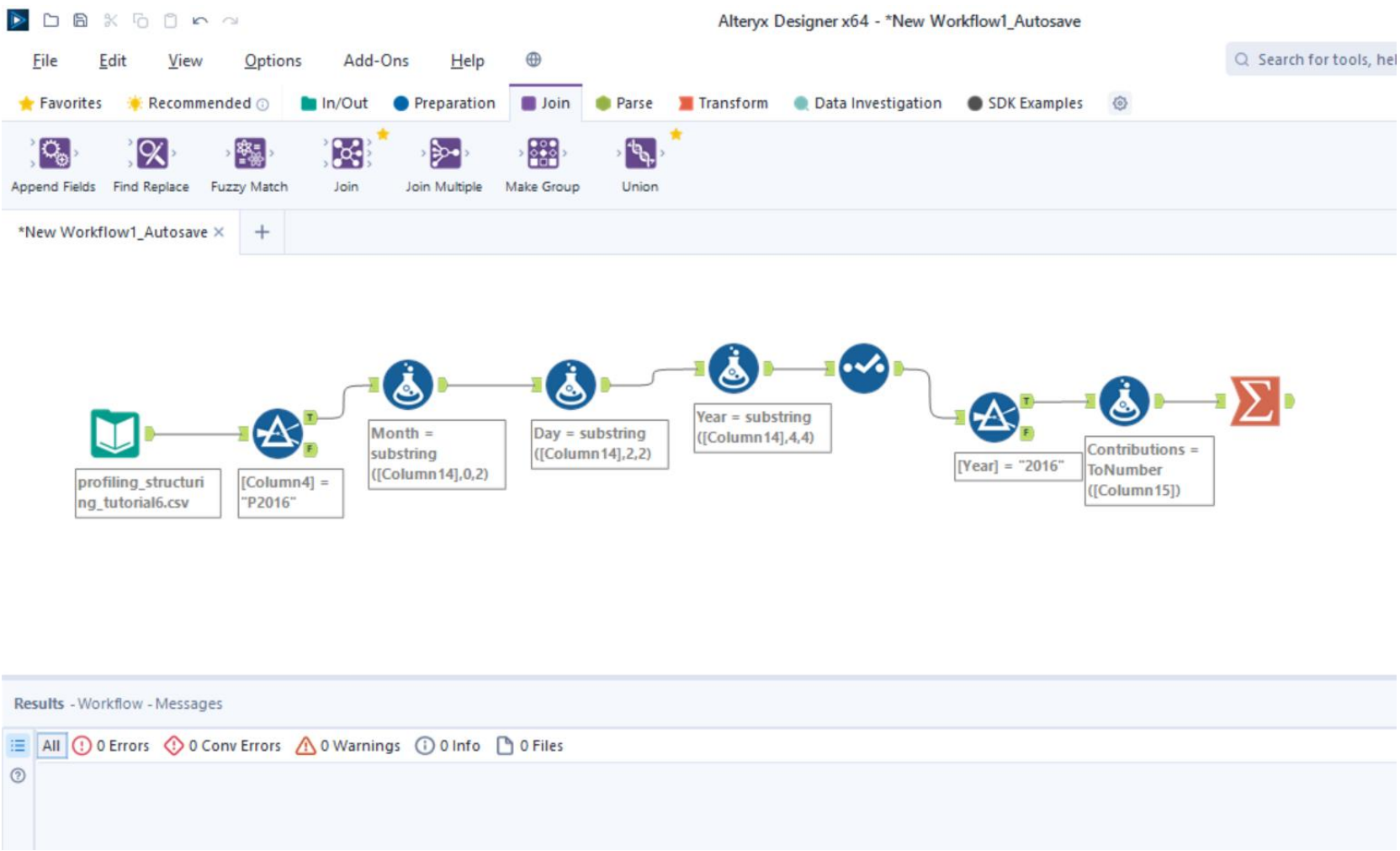

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Data Cleaning

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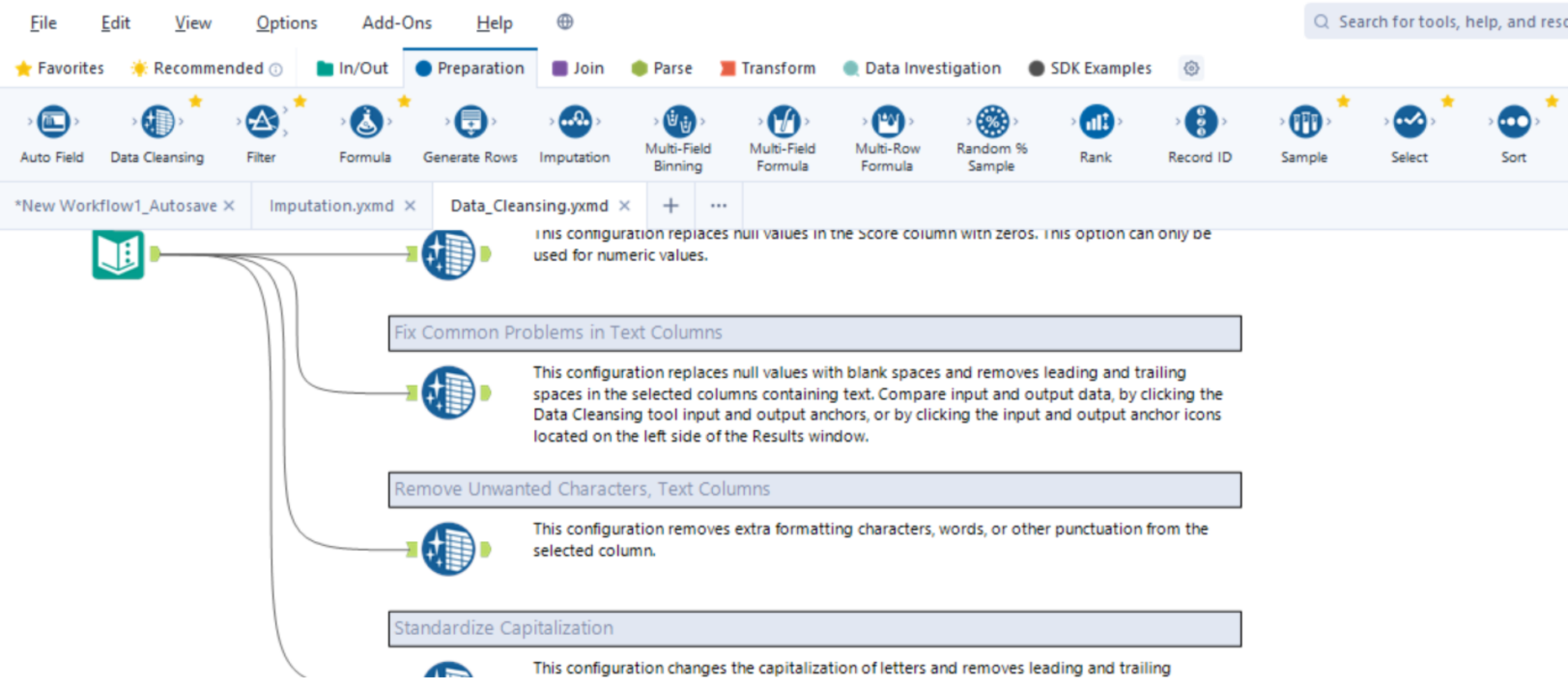

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Data cleaning overview

- Data cleaning is the process **of identifying and mitigating data errors** within a dataset.
- It is an essential step in data science that ensures that the **quality of data** is high to allow effective and reliable analysis and decision-making.
- **Data cleaning transformation actions** fall in two categories:
  - **Identification** of data quality issues
  - **Correction** or repair of the data

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Variety of Data Errors (a few examples)

representation

contradiction

referential integrity

CNr	Name	Birthday	Age	Sex	Phone	ZIP
1234	Smith, John	18.2.80	21	m	999-9999	70567
1234	Jane Doe	32.2.70	46	f	768-4511	55555
1235	John Smith	18.2.80	36	m	567-3211	70567

uniqueness

dummy value

duplicates

ZIP	Place
70567	Stuttgart
70567	Stutgart
70569	Germany

incorrect values

typo

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Variety of Data Errors (a few examples)

Errors that arise due to the **integration of overlapping, contradicting and inconsistent data** across sources.

- Contradictory values
- Different representations
- Different scales
- Different precision
- Duplicates
- Data conflicts
- Different level of aggregation

CID	Name	Size	Billing	Phone
1	Smith, John	8.5	80 USD	(123) 4567
2	Doe, Jane	7.0	70 USD	(408) 9012
A	John Smith	42	580 SGD	4444 8888
B	Chris Lim	42	420 SGD	9182 7374

Data from source 1

Data from source 2

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Cleaning actions overview

ERROR DETECTION
IDENTIFIES ERRONEOUS RECORDS OR ATTRIBUTE VALUES.

ERROR CORRECTION
CHANGES VALUES TO CORRECT ERRORS.

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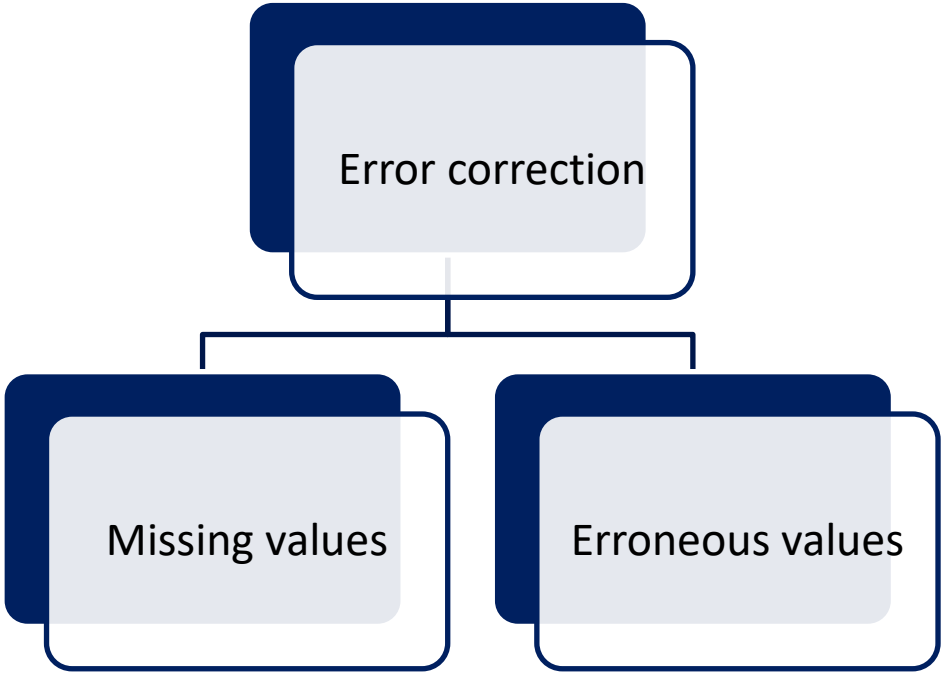
Error correction beyond deletion



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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik’s cube	10	NULL
01/01/2024	Bath towel	13	Home
01/01/2024	Rubik’s cube	NULL	Puzzle
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	NULL
01/01/2024	Bath towel	13	Home
01/01/2024	Rubik's cube	NULL	Puzzle
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Records complement each other and can be densified into one single tuple with no missing values

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	**Puzzle**
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	NULL	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Price is approximated based on the price of similar products (e.g., using the average)

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	**14**	NULL
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Category can be inferred by the category of similar products.

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	**14**	**Home**
03/01/2024	Large towel	15	Home
NULL	Jigsaw	20	Puzzle
05/01/2024	NULL	14	Home

Strong order of records allows to extrapolate possible date

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	**14**	**Home**
03/01/2024	Large towel	15	Home
04/01/2024	Jigsaw	20	Puzzle
05/01/2024	NULL	19	Home

Generative model can generate a likely value

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Correcting missing values

Date	Product	Price	Category
01/01/2024	Rubik's cube	10	Puzzle
01/01/2024	Bath towel	13	Home
02/01/2024	Beach towel	**14**	**HOME**
03/01/2024	Large towel	15	Home
04/01/2024	Jigsaw	20	Puzzle
05/01/2024	**XL towel**	19	Home

No more missing values

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	1.9	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
Zerez	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Standardization based on fixed library of valid names of solar system object. Replaced by most similar one.

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	1.9	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
Ceres	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Standardization using domain knowledge mapping planet names to their mass.

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	**189870.00**	Jovian planet	Gas
Uranus	8700	Jovian planet	Gas
Ceres	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Derivation of correctly formatted value (all others have two decimals).

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	Terrestril planet	Solid
Jupiter	**189870.00**	Jovian planet	Gas
Uranus	**8700.00**	Jovian planet	Gas
Ceres	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Correction of typographical error by replacing this outlier value by more frequent type (assume a larger table with few errors)

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	**Terrestrial planet**	Solid
Jupiter	**189870.00**	Jovian planet	Gas
Uranus	**8700.00**	Jovian planet	Gas
Ceres	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	Glace

Discovered functional dependency
Type → Surface
and value distribution allow to determine Ice as the correct value (assume a more complete relation)

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Correcting erroneous values

Object	Mass	Type	Surface
Earth	597.20	Terrestrial planet	Solid
Venus	486.70	**Terrestrial planet**	Solid
Jupiter	**189870.00**	Jovian planet	Gas
Uranus	**8700.00**	Jovian planet	Gas
Ceres	0.09	Dwarf planet	Ice
Pluto	1.30	Dwarf planet	**Ice**

No more erroneous values

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Data Cleaning comments

- Examples show value-specific corrections.
- A specific sequence of corrections may yield different results. Problematic?
- What data dependencies or constraints can be used?
- How to do this efficiently and effectively?
- ...

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Agenda

- Motivation and Introduction
- Common data preparation tasks
- Discussion phase: Example applications
- Wrap-up + Q&A

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Data Preparation experience in this room?

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Data preparation applications

Engineering

Architecture & Construction

Data preparation for AI

Manufacturing

Healthcare

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Share your stories

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To conclude

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Wrap-up

- Data preparation is essential when aiming at gaining insights from data.
- Data preparation is highly standardized with common types of operations.
- Advanced data preparation software can help reduce the effort needed for robust data preparation.

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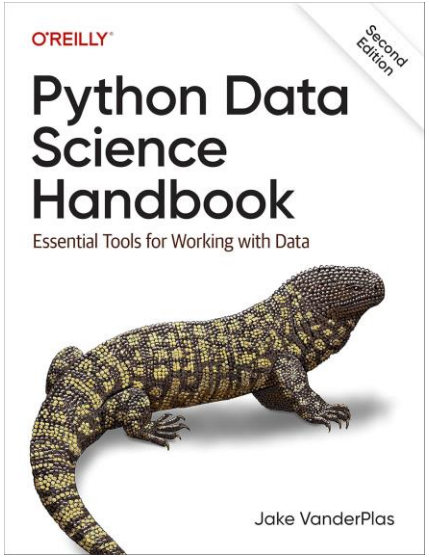
Useful Resources



General reference for coding:
Python Data Science Handbook. Jake VanderPlas, O'Reilly (2nd edition), 2023

- Talend Data Quality and Talend Data Preparation*
- IBM Infosphere Information Analyzer
- Oracle Enterprise Data Quality
- Microsoft Power Query
- Informatica Data Quality
- SAP Information Steward
- SAS Data Preparation*
- OpenRefine+
- Alteryx*
- Pandas profiling / ydata profiling+
- Great Expectations

** Means a trial version is available;*
+ means free / open-source version available

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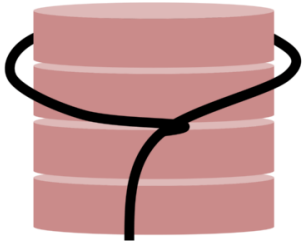
Pushing the limits of data preparation



Data integration and quality

Record linkage
Heterogeneous data
Adaptive algorithms

Data lakes

Accountable and fair data processing

Data provenance
Fairness and bias

Meta-data management

Trust in data engineering

Data exploration and analytics

Human-in-the-loop

Iterative pipeline refinement

Exploration and analytics assistants

Complex data pipelines

Pipeline debugging

Performance optimization

Domain-specific pipelines

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Thank you for your attention

Q&A

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